Qindong Gan

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EDUCATION

Waseda University **Tokyo, Japan**

Master of Engineering in Computer Science and Communications Engineering *09/2025-08/2027 (Expected)*

- **GPA: 3.33/4.00 (Waseda scale; A+ in 8 of 12 completed courses)**

University of Cincinnati **Cincinnati, OH, US**

Bachelor of Science in Electrical Engineering *09/2020-05/2025*

- **GPA: 3.571/4.000**

Joint program with Chongqing University; final academic year in Cincinnati.

Chongqing University **Chongqing, China**

Bachelor of Engineering in Electrical Engineering and Automation *09/2020-06/2025*

- **GPA: 3.35/4.00**

PUBLICATIONS

- **Q. Gan**, "When Self-Reported Fitness Becomes the Target: Independent Audits in LLM-Driven Recipe Evolution." **Submitted to** *Swarm and Evolutionary Computation* (SWEVO; JCR Q1). **Under review.**
- **Q. Gan**, Z. Pan, T. D. Ho, J. Wu, and S. Shimamoto, "Causally Verified QoS-Floor Constraints for Decentralized SAGIN Access Association," *IEEE Consumer Communications & Networking Conference (CCNC)*, 2027. **Accepted.**

PROJECTS

Constrained Multi-Agent RL for Space–Air–Ground Access Association [Link] **03/2026-08/2026**

First-author research (IEEE CCNC 2027) | Advisor: Shigeru Shimamoto, Waseda University

- **Problem:** Vehicles choose among LEO satellite, HAPS, and terrestrial tiers under co-tier interference; throughput-driven MARL meets long-run fairness while starving some vehicle in every slot. Built a 3GPP-based simulator and trained MAPPO (centralized training, decentralized execution) to protect a per-slot QoS floor.
- **Diagnosis:** Proved that the standard signed, fleet-relative constraint cannot drive a shared Lagrange multiplier upward and that per-vehicle expectation constraints are blind to intermittent starvation; traced an apparent 4.4× floor gain to an open-loop artifact of per-episode multiplier resets.
- **Innovation:** Proposed a causal test of whether a multiplier acts as feedback, cutting, misdirecting across agents, and sign-reversing its error signal; replaced the constraint with a one-sided expected-shortfall budget fixed before training from an untuned reference.
- **Results:** Raised the per-slot rate floor 17.1× (12 vehicles) and 63.1× (20 vehicles) over a matched unconstrained learner and cut outage at 0.1 Mbps from 35.0% to 7.2% for a 12.7% throughput cost, outperforming proportional-fair and greedy association on throughput, floor, and outage simultaneously.

Keywords: Constrained MARL; MAPPO; Lagrangian dual ascent; causal intervention; QoS fairness; SAGIN.

ShinsaEvolve: Independent Audits in LLM-Driven Recipe Evolution [Link] **09/2026-10/2026**

Independent research project; sole-author manuscript submitted to SWEVO

- **Problem:** LLM-driven program evolution selects candidates by self-reported scores; when each RL training recipe controls its own reward shaping and evaluation schedule, those scores lose a common scale and selection can reward the report rather than the policy.
- **Innovation:** Designed a harness-owned audit that scores only the trained policy parameters on 32 fresh seeds with raw environment reward, never executing candidate code; released an open, reproducible system with immutable run configurations, prompt hashing, and per-episode audit records.
- **Experiments:** Ran a 2×2 LunarLander study (audit vs. self-reported selection × candidate-shaped vs. fixed raw reward) with 12 outer-loop runs and 180 LLM-generated candidates trained by CMA-ES.
- **Results:** Under candidate-shaped reward, self-reported scores saturated and the selected policy averaged -762.1 audited return vs. 4.0 under audit selection; fixed raw reward narrowed the gap (-137.1 vs. -19.4) but misselection persisted in all three runs.

Keywords: LLM-driven program evolution; reinforcement learning; evaluation reliability; reward hacking; evolution strategies.

RGB-Thermal Semantic Segmentation for Autonomous Driving [Link] **07/2025-08/2025**

Full-time AI Algorithm Intern | Supervised by Jiachen Liang, SERES Group Co., Ltd.

- **Goal:** Pixel-level segmentation of driving scenes (e.g., road, vehicles, pedestrians) by fusing visible and thermal infrared images.
- **Approach:** Adapted DFormerV2's geometry-prior design, originally developed for RGB-D segmentation, to RGB-thermal input.
- **Work:** Contributed to project data collection; built, trained, and validated the RGB-thermal segmentation model.

Keywords: Multimodal semantic segmentation; RGB-thermal fusion; autonomous driving; DFormerV2.

RoboNeuro Mapping: Bio-inspired Visual Navigation for Robots [Link] **09/2024-05/2025**

Senior Capstone Project (team of three) | University of Cincinnati | Advisor: Ali Minai

- **Goal:** Extend a self-organizing hippocampal place-cell navigation model, previously driven by LiDAR alone, with camera input so that visual landmarks can disambiguate similar spatial contexts (Webots; iRobot Create 3).
- **Approach:** Embedded pretrained EfficientNetV2B0 visual features into the place-cell state space alongside LiDAR-based spatial representations for goal-directed navigation.
- **Work:** Collected and preprocessed simulation images, validated the image pipeline, and contributed to visual feature extraction and evaluation; coordinated team discussions and weekly progress reporting.

Keywords: Bio-inspired navigation; hippocampal place cells; visual–LiDAR fusion; EfficientNetV2; Webots.

Data-Driven Condition Monitoring of Power Electronics [Link] **05/2024-07/2024**

Research Assistant | Chongqing University | Advisor: Wei Lai

- **Goal:** Estimate capacitor aging condition robustly under measurement noise, sensor error, and changing operating conditions.
- **Work:** Ported the group's MATLAB sliding-window, wavelet time-frequency, and CNN-regression pipeline to Python with TensorFlow/Keras; integrated the model into LabVIEW via Python Node and tested it in simulation under noise and sensor errors.

Keywords: Condition monitoring; power electronics; time-frequency features; CNN regression; LabVIEW.

RESEARCH INTERESTS

Multi-agent reinforcement learning; decentralized decision-making; reliable policy evaluation.

HONORS AND AWARDS

- Best Master's Student Presentation Award, ASPIRE-CRC-TCA International Workshop, Memorial University, 2026.
- Dean's List, University of Cincinnati, Fall 2024 and Spring 2025.
- Cooperative Education & Practice - Excellence, University of Cincinnati.
- Second Prize, National College Student Robot Competition.
- Outstanding Student Scholarship (Top 10%); Outstanding Graduate, Chongqing University.

PRESENTATIONS AND TEACHING EXPERIENCE

Workshop Presentation **08/2026**

- "Multi-Agent Reinforcement Learning for Intelligent Access Management in Space-Air-Ground Integrated Networks." ASPIRE-CRC-TCA International Workshop, Memorial University.

Teaching Assistant, University of Cincinnati **09/2024-05/2025**

Courses: Foundations of Engineering Design Thinking I & II (ENED 1100 / ENED 1120).

- Supported undergraduate engineering instruction in English through recitations and labs; graded assignments and examinations.

SKILLS

Programming and tools: Python, MATLAB, PyTorch, TensorFlow/Keras, LabVIEW, EPyMARL, NumPy, pandas, scikit-learn, Git, LaTeX.

Methods: Multi-agent and constrained reinforcement learning, Lagrangian methods, simulation, ablation studies, multimodal perception.

Languages: Mandarin Chinese (native), English.